

Compendium on Neural Fields & NeRF architectures

A Technical Analysis from Coordinate-Based Models to Continuous Topological Maps

1. Introduction to Neural Radiance Fields (NeRF)

A **Neural Radiance Field (NeRF)** is a deep learning technology designed to **generate photorealistic 3D scenes and objects from a limited set of 2D images**. Introduced in 2020 by researchers from UC Berkeley, Google Research, and UC San Diego, it represents a profound paradigm shift away from traditional computer graphics. Instead of storing scene descriptions via explicit geometries like point clouds, meshes, or voxel grids, NeRF encapsulates the entire geometry and lighting of an environment **directly within the weights of a neural network**—specifically a Multi-Layer Perceptron (MLP).

How it works:

- **5D Input Vector:** For each spatial sample point, the network receives its 3D spatial coordinates (x, y, z) along with the camera's 2D viewing direction (θ, ϕ).
- **Model Output:** The MLP processes these five continuous variables to predict two core physical values: the volumetric density (σ) (opacity/translucency) and the emitted directional color **RGB**.
- **Ray Marching & Volume Rendering:** Differentiable volume rendering techniques simulate light transport by shooting virtual rays through pixels, sampling positions along these rays, and integrating colors and opacities to compute the final 2D image.
- **Optimization (Overfitting):** The network is trained by actively overfitting to a specific scene, minimizing the error between rendered pixels and the ground-truth input photographs.

2. Detailed Network Architecture Breakdown

The standard NeRF framework utilizes a surprisingly straightforward architecture: a fully connected Multi-Layer Perceptron (MLP) containing no convolutional blocks or attention layers. It acts strictly as a continuous function mapping coordinates to physical attributes.

- **Positional Encoding:** Neural networks exhibit a spectral bias, naturally favoring low-frequency functions and creating blurry details. To mitigate this, spatial coordinates and viewing directions are transformed into a higher-dimensional space using sine and cosine functions. Spatial coordinates (x, y, z) are expanded into 60 dimensions, while view directions are projected into 24 dimensions.
- **The Primary Density Block:** The first segment of the network takes *only* the positional coordinates (x, y, z) as input, since spatial opacity is view-independent. It consists of 8 dense layers, each containing 256 channels with ReLU activations. A skip connection concatenates the encoded input vector at the fifth layer to preserve spatial features. The block outputs the volume density (σ) and a 256-dimensional abstract geometric feature vector.

- **The Secondary Color Block:** The 256-dimensional feature vector is then concatenated with the encoded 24-dimensional view directions. Since surface reflection and highlights depend heavily on the viewing angle, this joined vector is processed by a secondary block consisting of a single dense layer of 128 channels. A final Sigmoid activation restricts output values to a strict [0, 1] range representing the final RGB channels.

3. The Concept of Neural Fields

NeRF is a highly successful manifestation of a broader computational philosophy known as a **Neural Field** (also termed *Implicit Neural Representation* or *Coordinate-Based Neural Network*). In physics, a field represents a continuous function that assigns a physical quantity to every point in a space-time domain. Analogously, a Neural Field uses a neural network to approximate a continuous function, accepting continuous space-time coordinates as inputs and outputting specific physical properties.

Field Paradigm	Input (Coordinates)	Output (Stored Property)	Resulting Representation
Neural Audio Field	Time (t)	Wave Amplitude	Continuous audio stream
Neural Image Field	Pixels (x, y)	Color (R, G, B)	Infinite-resolution image
Signed Distance Field (SDF)	Space (x, y, z)	Distance to nearest surface	Crisp geometric 3D mesh
NeRF (Neural Radiance Field)	Space + View (x, y, z, θ , ϕ)	Density & Radiance (σ , R, G, B)	Volumetric photorealistic 3D scene

4. Epistemological Debates: SOM vs. Continuous Mappings

A common question arises regarding whether classical connectionist architectures like **Self-Organizing Maps (SOM)** or *Kohonen Networks* qualify as Neural Fields. Standard SOMs generally do not fit the criteria. Standard SOMs operate as discrete vector quantization algorithms composed of a rigid, finite grid of nodes (e.g., a fixed 10x10 structure) mapping high-dimensional abstract input vectors to a discrete 'Winner-Take-All' index. They lack a continuous, differentiable spatial domain.

Bridging the Gap: Supervised Topological Maps (STM)

However, this distinction shifts fundamentally when examining extensions such as the **Supervised Topological Maps (STM)** proposed by F. Mannella. The STM framework transforms the discrete topological structure of self-organizing maps into a continuous domain through weighted interpolation mechanisms (often utilizing continuous Gaussian or normalized distance-based blending over the prototype node activations).

When deployed in a generative or reconstruction capacity, an STM effectively evaluates exact decimal coordinates within its map space to reconstruct full feature vectors: $f(\text{Map Coordinates}) \rightarrow \text{Feature Data}$. This mathematical formulation matches the operation of modern **Hybrid Neural Fields** (such as NVIDIA's Instant NGP or feature grid architectures), which marry discrete structural grids with continuous interpolation to yield

highly scalable neural representations. Consequently, continuous and interpolated variations of the SOM/STM lineage can legitimately be categorized as grid-backed topological Neural Fields.